

# Modern Mechanics Formulae and Data Sheet

## FUNDAMENTAL PHYSICAL LAWS

Principle of relativity: Physical laws work in the same way for observers in uniform motion as for observers at rest.  
The superposition principle: the effective force on an object is the “net” force, the vector sum of all forces acting on the object, each force unaffected by the presence of other interactions.

The momentum principle, and the definition of momentum. (These must be memorized.)

The energy principle, and the definitions of work and energy. (These must be memorized.)

The angular momentum principle. (This must be memorized.)

The relationship among position, velocity, and time. (This must be memorized.)

The second law of thermodynamics or the entropy principle: if a closed system is not in equilibrium, the most probable consequence is that the entropy will increase.

## MULTIPARTICLE SYSTEMS

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rel to cm}} = K_{\text{trans}} + K_{\text{rot}} + K_{\text{vib}} \quad K_{\text{total}} = \frac{1}{2} M_{\text{total}} v_{\text{cm}}^2 = \frac{P_{\text{total}}^2}{2M_{\text{total}}} \text{ nonrelativistically}$$

$$\vec{P}_{\text{total}} = M_{\text{total}} \vec{v}_{\text{cm}} \text{ nonrelativistically}$$

$$K_{\text{rot}} = \frac{L_{\text{rot}}^2}{2I} = \frac{1}{2} I \omega^2$$

$$I = m_1 r_{11}^2 + m_2 r_{22}^2 + m_3 r_{33}^2 + \dots$$

$$\vec{L}_A = \vec{r}_A \times \vec{p} \text{ for a point particle} \quad \vec{\tau}_A = \vec{r}_A \times \vec{F}$$

$$\vec{L}_A = \vec{L}_{\text{trans},A} + \vec{L}_{\text{rot}} = (\vec{R}_{\text{cm},A} \times \vec{P}_{\text{total}}) + \vec{L}_{\text{rot}} \text{ for a multiparticle system} \quad \vec{L}_{\text{rot}} = I \vec{\omega}$$

## EVALUATING GENERAL PHYSICAL QUANTITIES

$$E^2 - (pc)^2 = (mc^2)^2; \quad E = pc \text{ for massless photon}$$

$$\Omega = \frac{(q + N - 1)!}{q!(N - 1)!} \quad S = k \ln \Omega \quad \frac{1}{T} = \frac{dS}{dE} \quad \text{Stirling's approximation: } \ln n! \approx n \ln n - n$$

$$PV = (\text{number of moles})RT \quad R = N_A k_B$$

$$PV = \text{constant (for isotherm)} \quad PV^\gamma = \text{constant (for adiabat)} \quad \gamma = C_P / C_V$$

## EVALUATING SPECIFIC PHYSICAL QUANTITIES

$$|\vec{F}_{\text{grav}}| = G \frac{Mm}{|\vec{r}|^2}; \text{ can be approximately } mg$$

$$U_{\text{grav}} = \frac{-GMm}{|\vec{r}|}; \text{ change can be approximately } \Delta(mgy)$$

$$|\vec{F}_{\text{electric}}| = \frac{1}{4\pi\epsilon_0} \frac{|Qq|}{|\vec{r}|^2}$$

$$U_{\text{electric}} = \frac{1}{4\pi\epsilon_0} \frac{Qq}{|\vec{r}|}$$

$$|\vec{F}_{\text{spring}}| = k_s |s|, \text{ opposite the stretch}$$

$$U_{\text{spring}} = \frac{1}{2} k_s s^2 + U_0$$

$$\vec{F} = \left\langle -\frac{\partial U}{\partial x}, -\frac{\partial U}{\partial y}, -\frac{\partial U}{\partial z} \right\rangle$$

$$\text{Power} = \text{energy/time (watts = joules/second)}$$

$$\text{Perpendicular component of } d\vec{p}/dt \text{ (toward the centre of the kissing circle): } \left| \frac{\vec{v}}{R} \right| |\vec{p}|$$

$$Y = \frac{F/A}{\Delta L/L}$$

$$|\vec{v}| = \frac{2\pi|\vec{r}|}{T} = \omega|\vec{r}|$$

$$x = A \cos \left( \sqrt{\frac{k_s}{m}} t \right)$$

$$\text{air resistance: } F = -\frac{1}{2} C \rho A v^2$$

$$\text{sliding friction: } f_s \leq \mu_s F_N$$

$$\text{oscillator: } E_N = \left( N + \frac{1}{2} \right) \left( \hbar \sqrt{\frac{k_s}{m}} \right); \quad \text{classically } \omega = \sqrt{\frac{k_s}{m}}$$

$$\text{hydrogen atom: } E_N = -\frac{13.6 \text{ eV}}{N^2} \quad (N = 1, 2, \dots)$$

Macro/micro connections:

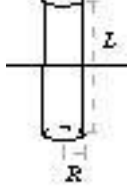
$$\text{macro measurement of density} = \frac{m}{d^3} \quad (\text{micro quantities})$$

$$\text{macro measurement of Young's modulus } Y = \frac{k_{s,\text{interatomic}}}{d_{\text{atomic}}} \quad (\text{micro quantities})$$

$$\text{macro measurement of speed of sound } v = d \sqrt{\frac{k_s}{m}} \quad (\text{micro quantities})$$

$$C = \frac{\Delta E}{\Delta T} \quad (\text{high-temperature limit: } \frac{1}{2} k_B \text{ per quadratic energy term; solid is } 3k_B \text{ per atom})$$

$$\text{Boltzmann } P(\Delta E) = \exp\left\{-\frac{\Delta E}{k_B T}\right\} \quad \text{Maxwell-Boltzmann } P(v)dv = 4\pi \left(\frac{M}{2\pi k_B T}\right)^{\frac{3}{2}} \exp\left\{-\frac{\frac{1}{2} M v^2}{k_B T}\right\} v^2 dv$$

| Sphere                                                                            | Cylinder or disk                                                                  | Thin rod (about axis shown)                                                       | Solid cylinder (about axis shown)                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| $I = \frac{2}{5} MR^2$                                                            | $I = \frac{1}{2} MR^2$                                                            | $I = \frac{1}{12} ML^2$                                                           | $I = \frac{1}{12} ML^2 + \frac{1}{4} MR^2$                                          |

### CONSTANTS

$$G = 6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

$$g = 9.8 \text{ m s}^{-2}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

$$h = 6.6 \times 10^{-34} \text{ J s}$$

$$\hbar = 1.05 \times 10^{-34} \text{ J s}$$

$$k_B = 1.4 \times 10^{-23} \text{ J K}^{-1}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$$

$$\text{proton or positron charge: } e = 1.6 \times 10^{-19} \text{ coulomb}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ molecules/mole}$$

$$m_{\text{proton}} \approx m_{\text{neutron}} \approx m_{\text{hydrogen atom}} = \frac{1 \times 10^{-3} \text{ kg/mole}}{6 \times 10^{23} \text{ atoms/mole}} = 1.7 \times 10^{-27} \text{ kg}$$

$$m_{\text{electron}} = 9 \times 10^{-31} \text{ kg}$$

$$M_{\text{Earth}} = 6 \times 10^{24} \text{ kg}$$

$$\text{Radius of Earth} = 6.4 \times 10^6 \text{ m}$$

$$M_{\text{Moon}} = 7 \times 10^{22} \text{ kg}$$

$$\text{Distance from Earth to Moon} = 4 \times 10^8 \text{ m}$$

$$M_{\text{Sun}} = 2 \times 10^{30} \text{ kg}$$

$$\text{Distance from Earth to Sun} = 1.5 \times 10^{11} \text{ m}$$

$$\text{Typical atomic radius } r \approx 1 \times 10^{-10} \text{ m}$$

$$\text{Proton radius } r \approx 1 \times 10^{-15} \text{ m, } R_{\text{nucleus}} \approx (1.3 \times 10^{-15} \text{ m}) A^{1/3}$$

$$\text{Heat capacity of water} = 4.2 \text{ (J/K)/gram.}$$

$$\text{Energy of visible light is from about 1.8 eV to about 3.1 eV.}$$

$$\text{Effective stiffness of interatomic bond in aluminium is } 16 \text{ N m}^{-1}, \quad \text{lead is } 5 \text{ N/m.}$$

$$\text{Density of water} = 1000 \text{ kg m}^{-3}$$

Mass of one mole of *atoms*:

$$\text{carbon 12 grams (0.012 kg)}$$

$$\text{nitrogen 14 grams (0.014 kg)}$$

$$\text{oxygen 16 grams (0.016 kg)}$$

$$\text{aluminum 27 grams (0.027 kg)}$$

$$\text{iron 56 grams (0.056 kg)}$$

$$\text{lead 207 grams (0.207 kg)}$$